

IF440: Artificial Intelligence

Topic 11: Fuzzy Logic

Moeljono Widjaja, Ph.D.
Yaya Suryana, Ph.D.

Universitas Multimedia Nusantara
Jl. Scientia Boulevard, Gading Serpong
Tangerang, Banten-15811 Indonesia

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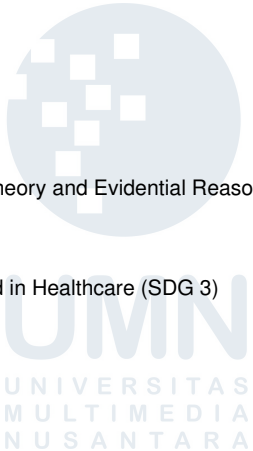
Course Sub Learning Outcomes (Sub-CLO):



SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).

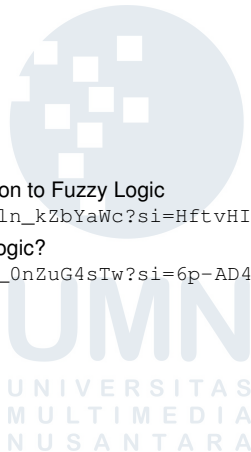


Review

- 
- 1 Bias of Bayesian Method
 - 2 Certainty Factors (CF) Theory and Evidential Reasoning
 - 3 Example of CFs Theory
 - 4 Dempster-Shafer Theory
 - 5 Application of DS Method in Healthcare (SDG 3)
 - 6 Exercises

Pre-class Assignment

- Tim Arnett: An Introduction to Fuzzy Logic
https://youtu.be/rln_kZbYaWc?si=HftvHIX_31GhRe2a
- Matlab: What is Fuzzy Logic?
https://youtu.be/___0nZuG4sTw?si=6p-AD4SwfhgaGy3L



Outline

- 1 Introduction to Fuzzy Logic
 - Crisp Sets
 - Fuzzy Set Theory
 - Operations on Fuzzy Sets
 - Membership Functions

- 2 Fuzzy Inference System
 - Fuzzification
 - Defuzzification
 - Fuzzy Rules

- 3 Exercise



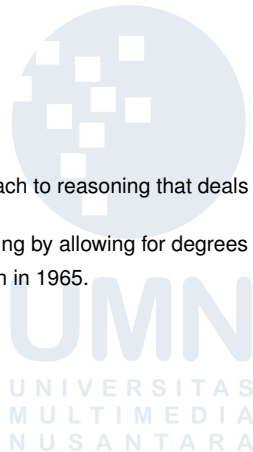


Introduction to Fuzzy Logic

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What is Fuzzy Logic?

- Fuzzy Logic is an approach to reasoning that deals with approximate rather than fixed and exact values.
- It models human reasoning by allowing for degrees of truth.
- Originated by Lotfi Zadeh in 1965.



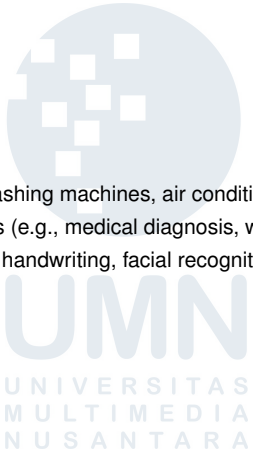
Fuzzy Logic vs. Classical Logic

Table: Comparison of Classical and Fuzzy Logic

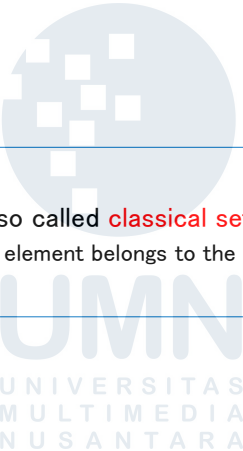
Aspect	Classical Logic	Fuzzy Logic
Truth values	Binary (0 or 1)	Continuous (0 to 1)
Precision	Exact	Approximate
Examples	Digital circuits	Control systems

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Applications of Fuzzy Logic

- 
- Control systems (e.g., washing machines, air conditioners).
 - Decision-making systems (e.g., medical diagnosis, weather prediction).
 - Pattern recognition (e.g., handwriting, facial recognition).

CRISP SETS

A large, light blue watermark of the Universitas Multimedia Nusantara (UMN) logo is centered in the background. It features a circular emblem with a grid of squares and the text 'UMN' and 'UNIVERSITAS MULTIMEDIA NUSANTARA' below it.

Also called **classical sets**

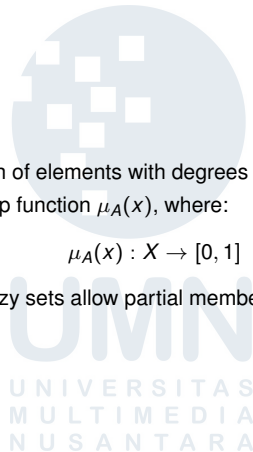
- either an element belongs to the set or not.

What is a Fuzzy Set?

- A fuzzy set is a collection of elements with degrees of membership.
- Defined by a membership function $\mu_A(x)$, where:

$$\mu_A(x) : X \rightarrow [0, 1]$$

- Unlike classical sets, fuzzy sets allow partial membership.

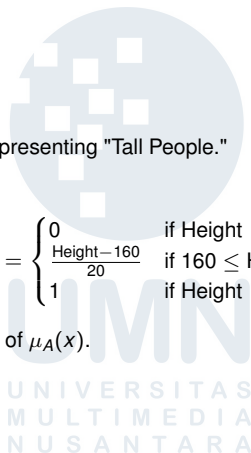


Example of a Membership Function

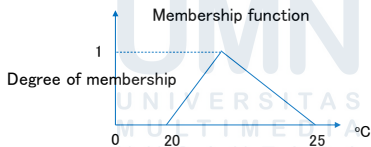
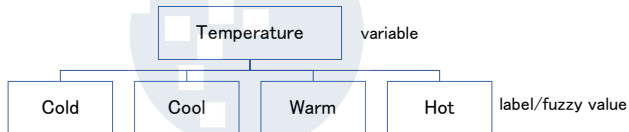
- Example: Fuzzy set A representing "Tall People."
- Membership function:

$$\mu_A(\text{Height}) = \begin{cases} 0 & \text{if Height} < 160 \text{ cm} \\ \frac{\text{Height} - 160}{20} & \text{if } 160 \leq \text{Height} \leq 180 \text{ cm} \\ 1 & \text{if Height} > 180 \text{ cm} \end{cases}$$

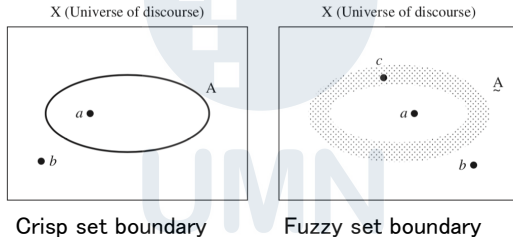
- Graphical representation of $\mu_A(x)$.



FUZZY SET THEORY



FUZZY SET THEORY



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Operations on Fuzzy Sets

■ Union ($A \cup B$):

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

■ Intersection ($A \cap B$):

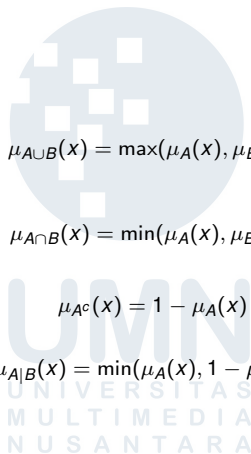
$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

■ Complement (A^c):

$$\mu_{A^c}(x) = 1 - \mu_A(x)$$

■ Difference ($A|B$):

$$\mu_{A|B}(x) = \min(\mu_A(x), 1 - \mu_B(x))$$



OPERATIONS ON FUZZY SET

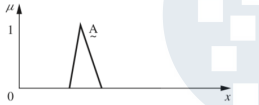


FIGURE 2.11
Membership function for fuzzy set $\underline{A}$.

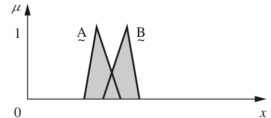


FIGURE 2.12
Union of fuzzy sets $\underline{A}$ and $\underline{B}$.

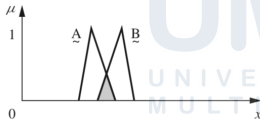


FIGURE 2.13
Intersection of fuzzy sets $\underline{A}$ and $\underline{B}$.

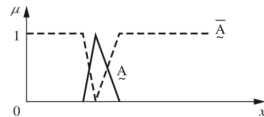


FIGURE 2.14
Complement of fuzzy set $\underline{A}$.

OPERATIONS ON FUZZY SET

Example: $U = \{1, 2, 3, 4, 5\}$

$$\underline{A} = \left\{ \frac{1}{2} + \frac{0.5}{3} + \frac{0.3}{4} + \frac{0.2}{5} \right\}$$

$$\underline{B} = \left\{ \frac{0.5}{2} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.4}{5} \right\}$$

Find the union, intersection, complement, and difference!

OPERATIONS ON FUZZY SET

$$\underline{\underline{A}} = \left\{ \frac{1}{2} + \frac{0.5}{3} + \frac{0.3}{4} + \frac{0.2}{5} \right\} \quad \text{and} \quad \underline{\underline{B}} = \left\{ \frac{0.5}{2} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.4}{5} \right\}$$

Complement

$$\overline{\underline{\underline{A}}} = \left\{ \frac{1}{1} + \frac{0}{2} + \frac{0.5}{3} + \frac{0.7}{4} + \frac{0.8}{5} \right\}.$$

$$\overline{\underline{\underline{B}}} = \left\{ \frac{1}{1} + \frac{0.5}{2} + \frac{0.3}{3} + \frac{0.8}{4} + \frac{0.6}{5} \right\}.$$

Union

$$\underline{\underline{A}} \cup \underline{\underline{B}} = \left\{ \frac{1}{2} + \frac{0.7}{3} + \frac{0.3}{4} + \frac{0.4}{5} \right\}.$$

Intersection

$$\underline{\underline{A}} \cap \underline{\underline{B}} = \left\{ \frac{0.5}{2} + \frac{0.5}{3} + \frac{0.2}{4} + \frac{0.2}{5} \right\}.$$

Difference

$$\underline{\underline{A}} \setminus \underline{\underline{B}} = \underline{\underline{A}} \cap \overline{\underline{\underline{B}}} = \left\{ \frac{0.5}{2} + \frac{0.3}{3} + \frac{0.3}{4} + \frac{0.2}{5} \right\}.$$

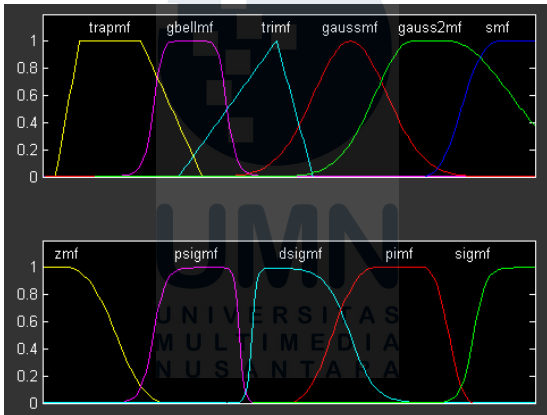
$$\underline{\underline{B}} \setminus \underline{\underline{A}} = \underline{\underline{B}} \cap \overline{\underline{\underline{A}}} = \left\{ \frac{0}{2} + \frac{0.5}{3} + \frac{0.2}{4} + \frac{0.4}{5} \right\}.$$

Membership Functions

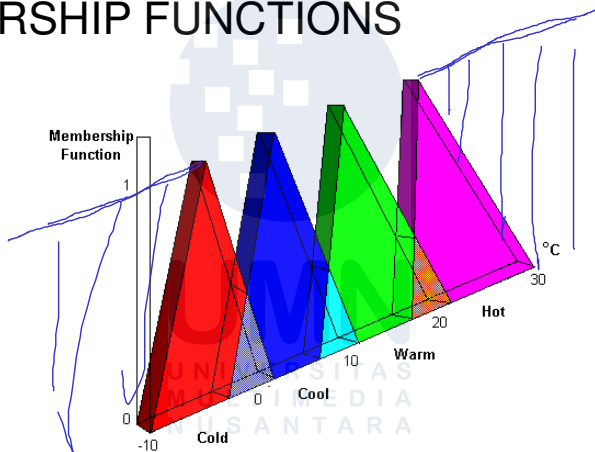
- Represent how each point in the input space is mapped to a membership value μ between 0 and 1.
- Common types:
 - Triangular
 - Trapezoidal
 - Gaussian
 - Sigmoidal



MEMBERSHIP FUNCTIONS



MEMBERSHIP FUNCTIONS



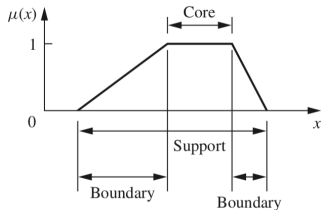
FEATURES OF MEMBERSHIP FUNCTIONS

- **Support of a fuzzy set**

- The support of $A \rightarrow \text{Supp}(A)$.
- Crisp subset of U which contains all elements of U that **have a non-zero membership** value in A .

- **Core of a fuzzy set**

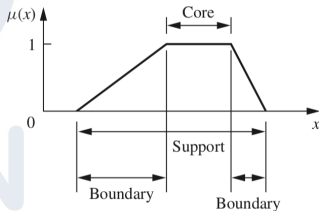
- The core of $A \rightarrow \text{Core}(A)$.
- Crisp subset of U which consists all elements with **membership grade 1**.



FEATURES OF MEMBERSHIP FUNCTIONS

- **Boundaries of a fuzzy set**

- The elements x of the universe such that $0 < \mu_A(x) < 1$



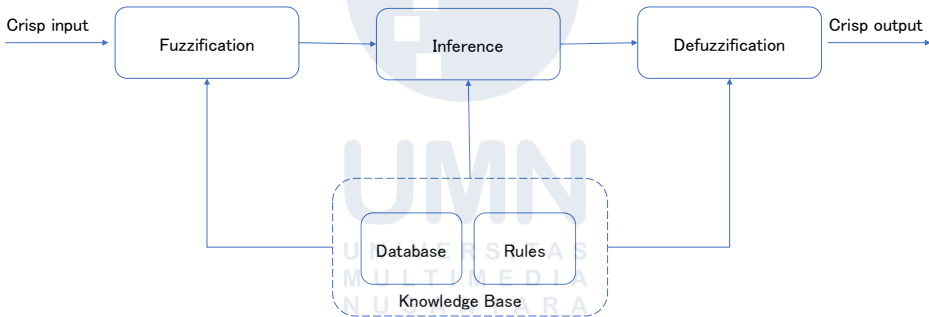
Fuzzy Inference System



FUZZY INFERENCE SYSTEMS

A system that uses fuzzy set theory to map inputs to outputs

FUZZY INFERENCE SYSTEMS



What is Fuzzification?

- 
- Fuzzification is the process of transforming **crisp inputs** into **fuzzy values**.
 - It maps real-world data to degrees of membership in predefined fuzzy sets.
 - This is the first step in a fuzzy inference system.

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Why is Fuzzification Important?

- Allows handling of **uncertainty** and **imprecision**.
- Enables **approximate reasoning** like human decision-making.
- Crucial for applications where precise data is unavailable or unreliable.

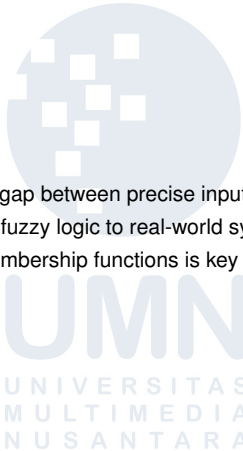


Fuzzification Output

- 
- A diagram illustrating the fuzzification process. It features a large light blue circle containing several smaller white squares. Some of these squares are solid white, while others are semi-transparent, representing different degrees of membership in a fuzzy set.
- Output of fuzzification is a set of **degrees of membership**.
 - Each input can belong to multiple fuzzy sets with varying degrees.
 - This output is passed to the **inference engine** in fuzzy systems.

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Remarks on Fuzzification

- 
- Fuzzification bridges the gap between precise input and fuzzy reasoning.
 - It's essential for applying fuzzy logic to real-world systems.
 - Selecting appropriate membership functions is key for accuracy.

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FUZZIFICATION

Process of changing a real scalar value into a fuzzy value using
membership functions stored in knowledge base

FUZZIFICATION

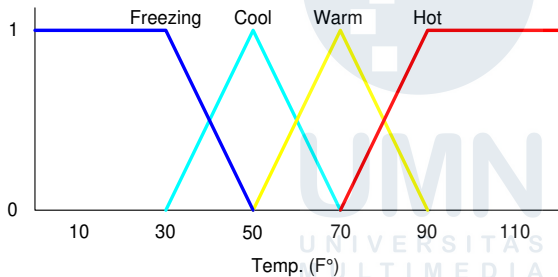
What is the temperature?

How warm is it?

It is warm

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FUZZIFICATION



How cool is 36°F ?

It is 30% Cool and 70% Freezing

Worked Example: Fuzzification

Input Variable: Temperature in °C

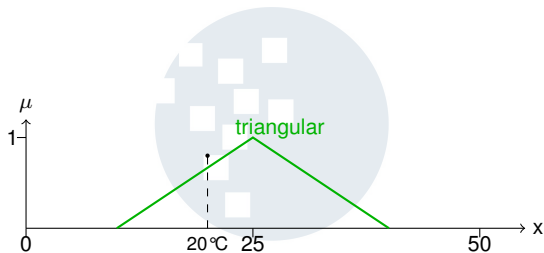
Fuzzy Sets (Triangular, Overlapping):

- Low: (0, 0, 25)
- Medium: (0, 25, 50)
- High: (25, 50, 50)

Input Value: 20°C



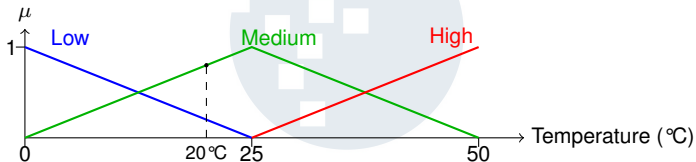
Triangular Membership Function Formula



$$\mu(x) = \begin{cases} 0 & x \leq a \text{ or } x \geq c \\ \frac{x-a}{b-a} & a < x < b \\ \frac{c-x}{c-b} & b \leq x < c \end{cases}$$

Where a = left, b = peak, c = right of triangle.

Membership in *Low* (0, 0, 25)

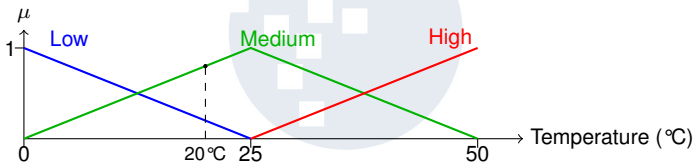


$$x = 20, \quad a = 0, \quad b = 0, \quad c = 25$$

Since $b \leq x < c$:

$$\mu_{\text{Low}}(20) = \frac{25 - 20}{25 - 0} = \frac{5}{25} = 0.2$$

Membership in *Medium* (0, 25, 50)

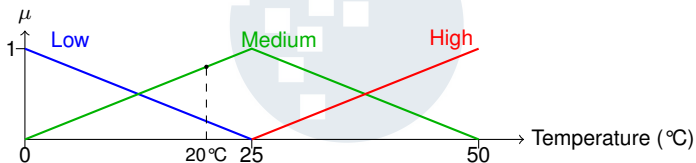


$$x = 20, \quad a = 0, \quad b = 25, \quad c = 50$$

Since $a < x < b$:

$$\mu_{\text{Medium}}(20) = \frac{20 - 0}{25 - 0} = \frac{20}{25} = 0.8$$

Membership in *High* (25, 50, 50)



$$x = 20, \quad a = 25, \quad b = 50, \quad c = 50$$

Since $x < a$:

$$\mu_{\text{High}}(20) = 0$$

Fuzzified Output Summary

At $x = 20^{\circ}\text{C}$, the degrees of membership are:

- **Low:** $\mu_{\text{low}} = 0.2$
- **Medium:** $\mu_{\text{medium}} = 0.8$
- **High:** $\mu_{\text{high}} = 0$

Interpretation: Temperature is mostly *Medium*, slightly *Low*.



What is Defuzzification?



- Defuzzification converts a fuzzy output set into a crisp value.
- Essential in fuzzy systems to make real-world decisions.

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DEFUZZIFICATION

Process of changing a fuzzy value into a real scalar value

DEFUZZIFICATION

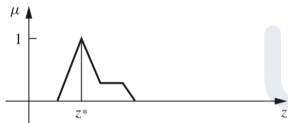
1. Max Membership Principle
2. Centroid of Area (COA) / Center of Gravity (COG)
3. Weighted Average
4. Center of Largest Area / Bisector of Area (BOA)
5. Center of Sums (COS)
6. Mean Max Membership
7. First of Max (and Last of Max)



DEFUZZIFICATION

1. Max Membership Principle

$$\mu_{\zeta}(z^*) \geq \mu_{\zeta}(z), \quad \text{for all } z \in Z,$$

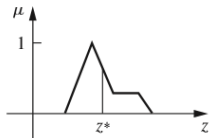


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DEFUZZIFICATION

2. Centroid of Area (COA) / Center of Gravity (COG)

$$z^* = \frac{\int \mu_{\zeta}(z) \cdot z \, dz}{\int \mu_{\zeta}(z) \, dz},$$

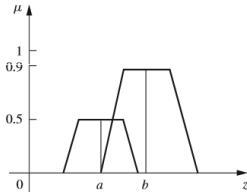


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DEFUZZIFICATION

3. Weighted Average

$$z^* = \frac{\sum \mu_{\tilde{C}}(\bar{z}) \cdot \bar{z}}{\sum \mu_{\tilde{C}}(\bar{z})},$$



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What is a Fuzzy Rule?

- 
- A fuzzy rule is an **IF-THEN** statement used in fuzzy inference systems.
 - **Format:** IF x is A THEN y is B
 - **Example:** IF temperature is high THEN fan speed is fast

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Multi-Condition Rules

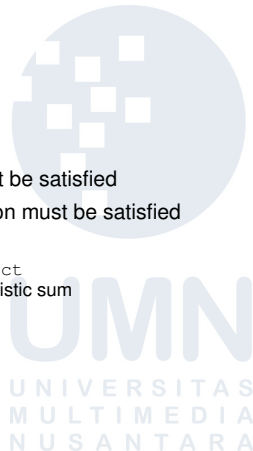
- Rules may involve multiple conditions:

- IF temperature is high AND humidity is low THEN fan speed is fast
- IF temperature is high OR humidity is high THEN fan speed is fast



Connectors: AND / OR

- **AND:** All conditions must be satisfied
- **OR:** At least one condition must be satisfied
- Fuzzy operators:
 - AND $\rightarrow$ min or product
 - OR $\rightarrow$ max or probabilistic sum

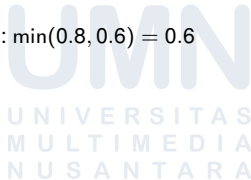


Rule Evaluation Example (AND)

Rule:

IF temperature is high AND humidity is low THEN fan speed is fast

- $\mu_{\text{high}}(\text{temperature}) = 0.8$
- $\mu_{\text{low}}(\text{humidity}) = 0.6$
- Combined degree (AND): $\min(0.8, 0.6) = 0.6$



Rule Evaluation Example (OR)

Rule:

IF temperature is high OR humidity is high THEN fan speed is fast

- $\mu_{\text{high}}(\text{temperature}) = 0.8$
- $\mu_{\text{high}}(\text{humidity}) = 0.6$
- Combined degree (OR): $\max(0.8, 0.6) = 0.8$



AND using Product (T-norm)

Rule: IF temperature is high AND humidity is low THEN fan speed is fast

- $\mu_{\text{high}}(T) = 0.8, \mu_{\text{low}}(H) = 0.6$
- Combined degree (AND - product): $0.8 \times 0.6 = 0.48$

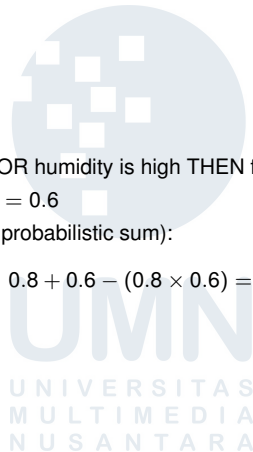


OR using Probabilistic Sum (S-norm)

Rule: IF temperature is high OR humidity is high THEN fan speed is fast

- $\mu_{\text{high}}(T) = 0.8, \mu_{\text{high}}(H) = 0.6$
- Combined degree (OR - probabilistic sum):

$$0.8 + 0.6 - (0.8 \times 0.6) = 0.92$$



Comparison of AND vs OR

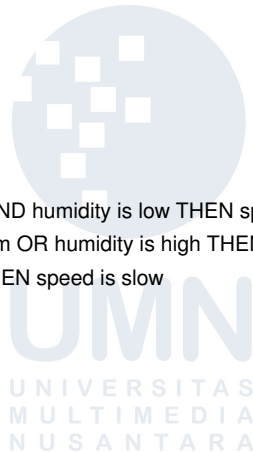
Connector	Operation	Result	Use Case
AND	min	Conservative	All conditions must be true
AND	$\cdot$ (product)	Flexible	When partial satisfaction is acceptable
OR	max	Optimistic	At least one condition true
OR	Prob. sum	Enhanced flexibility	Aggregated influence from both

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Application Example

Fan Controller Rules:

- 1 IF temperature is high AND humidity is low THEN speed is fast
- 2 IF temperature is medium OR humidity is high THEN speed is medium
- 3 IF temperature is low THEN speed is slow



Remarks on Fuzzy Rules

- 
- Fuzzy rules can use AND/OR to combine multiple premises.
 - AND $\rightarrow$ use min, OR $\rightarrow$ use max.
 - Useful for modeling complex decision systems.

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Fuzzy Inference Models

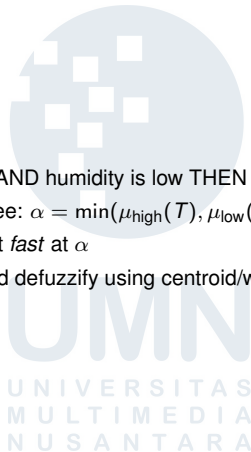
- **Mamdani:** outputs fuzzy sets; uses centroid/weighted-average defuzzification.
- **Sugeno:** outputs crisp values as linear functions or constants.
- **Tsukamoto:** outputs crisp values from monotonic membership functions.



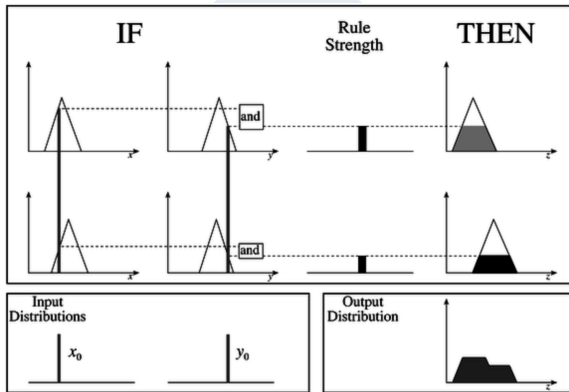
Mamdani Example

Rule: IF temperature is high AND humidity is low THEN fan speed is fast

- Compute minimum degree: $\alpha = \min(\mu_{\text{high}}(T), \mu_{\text{low}}(H))$
- Truncate output fuzzy set *fast* at α
- Aggregate all outputs and defuzzify using centroid/weighted-average method



Fuzzy Inference Systems (Mamdani)



A two input, two rule Mamdani FIS with crisp inputs

Source: <https://www.cs.princeton.edu/courses/archive/fall07/cos436/HIDDEN/Knapp/fuzzy004.htm>

htm

Sugeno Example

Rule: IF temperature is high AND humidity is low THEN fan speed = $0.5T + 0.3H + 2$

- Compute firing strength: $\alpha = \min(\mu_{\text{high}}(T), \mu_{\text{low}}(H))$
- Output is weighted average of all rule outputs
- Result is a crisp value



FIS (SUGENO)

IF x is A and y is B THEN z is $f(x,y)$ → Crisp function

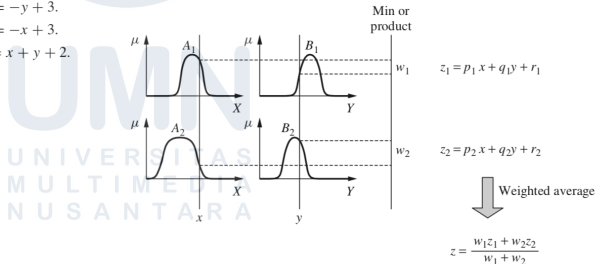
Example 5.16. An example of a two-input, single-output Sugeno model with four rules is repeated from Jang *et al.* (1997):

IF X is small and Y is small, THEN $z = -x + y + 1$.

IF X is small and Y is large, THEN $z = -y + 3$.

IF X is large and Y is small, THEN $z = -x + 3$.

IF X is large and Y is large, THEN $z = x + y + 2$.



Tsukamoto Example

Rule: IF temperature is high THEN fan speed is slow (monotonic fuzzy set)

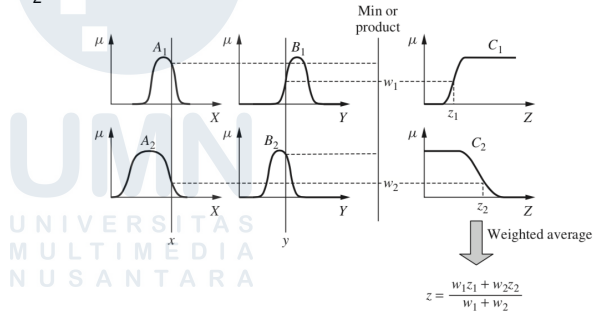
- Compute $\mu_{\text{high}}(T)$
- Apply inverse of output membership function to get crisp output
- Combine outputs using weighted average



FIS (TSUKAMOTO)

IF x is A_1 and y is B_1 THEN z is C_1

IF x is A_2 and y is B_2 THEN z is C_2





Exercise

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EXERCISE

Fan Rotation Frequency Control System

- The system control is affected by **speed of rotation**, **room temperature**, and resource of **rotation frequency**.
- Based on specification data from factory, the slowest speed is 1000rpm and the fastest speed is 5000rpm. The temperature is from 100K – 600K. The frequency is from 2000rpm – 7000rpm.

[R1] IF speed is slow and temperature is high THEN frequency is small.

[R2] IF speed is slow and temperature is low THEN frequency is small.

[R3] IF speed is fast and temperature is high THEN frequency is large.

[R4] IF speed is fast and temperature is low THEN frequency is large.

Q: What is the output, if the temperature is 300K and the speed is 4000rpm? With **Mamdani and Tsukamoto**

EXERCISE

Fan Rotation Frequency Control System

- Rule changes for Sugeno:

[R1] IF speed is slow and temperature is high THEN frequency = $0.5 * \text{speed} + 1700$.

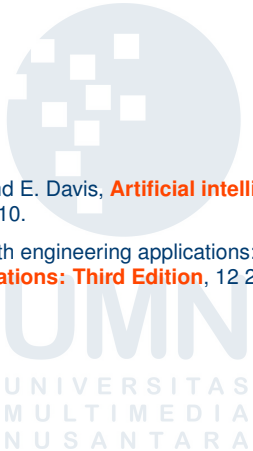
[R2] IF speed is slow and temperature is low THEN frequency = $2 * \text{speed} - 4000$.

[R3] IF speed is fast and temperature is high THEN frequency = $0.5 * \text{speed} + 2000$.

[R4] IF speed is fast and temperature is low THEN frequency = $\text{speed} + 700$.

Q: What is the output, if the temperature is 300K and the speed is 4000rpm? **With Sugeno.**

References

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- [1] S. J. Russell, P. Norvig, and E. Davis, **Artificial intelligence: a modern approach**, 3rd ed. Prentice Hall, 2010.
- [2] T. J. Ross, "Fuzzy logic with engineering applications: Third edition," **Fuzzy Logic with Engineering Applications: Third Edition**, 12 2010.

Next Week

- Introduction to Genetic Algorithm
- Chromosome
- Parent Selection
- Crossover
- Mutation
- Exercise



Vision and Mission

Vision

Becoming a leading Informatics Undergraduate Study Program that produces graduates with international insight who are competent in the field of computer science, have an entrepreneurial spirit, and have noble character.



Mission

- 1 Organising learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.

